

Project Proposal

Course: INTELLIGENT SYSTEMS

Title: Predicting Air Quality in Italy Using Fuzzy and Neuro-Fuzzy Modelling

Group Member:

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Motivation

Air quality prediction is vital for understanding pollution dynamics and public health impacts. Real-world sensor data are affected by noise and drift, making classical regression unreliable. Fuzzy logic provides a way to handle uncertainty through linguistic rules. This project applies fuzzy and neuro-fuzzy systems to estimate pollutant concentrations from raw sensor data, emphasizing adaptive and time-aware modeling.

Objectives

- Preprocess and analyze the Air Quality dataset, addressing missing data, scaling, and drift.
- Develop fuzzy models for pollutant prediction using fuzzy clustering for rule generation.
- Implement a neuro-fuzzy architecture (e.g., ANFIS) capable of adapting to nonlinear and time-dependent behavior.
- Explore deep models such as LSTM and CNN to capture temporal and local patterns.
- Compare all approaches (fuzzy, neuro-fuzzy, deep learning) with baseline regressors using error metrics.

Dataset

- **Source:** *Air Quality Dataset*, UCI Machine Learning Repository. <https://archive.ics.uci.edu/dataset/360/air+quality>

- **Description:** 9,358 hourly records (Mar 2004–Feb 2005) with 15 numerical features from five metal-oxide sensors, meteorological variables, and reference pollutant concentrations (CO, NOx, NO₂, benzene). Missing values denoted by -200.

Remarks

Implementation will use **Python** with NumPy, pandas, scikit-fuzzy, and TensorFlow. Evaluation will rely on RMSE, MAE, and R². Main challenges include managing drift, tuning membership functions, and balancing interpretability with accuracy.